OPTIMISATION, CONTROL, AND DATA

M2 Mathématiques & Applications (Sorbonne Université)

From gradient descent to optimal control

Numerical methods · Controllability · Linear-quadratic problems

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Slides for chapters 3–5 of the lecture notes · 2026

Outline

These slides cover chapters 3 to 5 of the lecture notes. Numbered results match the notes.

Chapter 3 **Numerical methods for (non)convex optimization.**
Gradient flow and gradient descent, with convergence rates. Stochastic gradient descent (results without proofs). Proximal and dual methods for non-smooth problems.

Chapter 4 **Controllability.**
Control systems, the end-point map and reachable sets. The Kalman rank condition and the Hautus test. Normal forms. Time-varying systems and the controllability Gramian.

Chapter 5 **Optimal control theory for linear systems.**
Linear-quadratic problems: existence, the Pontryagin Maximum Principle, its reading as a KKT condition, the Riccati equation, and the tracking problem.

3

chapter

Numerical methods for (non)convex optimization

- Gradient descent
- Stochastic gradient descent
- Proximal methods
- Dual methods

3.1

Numerical methods for (non)convex optimization

Gradient descent

The problem, and the simplest idea

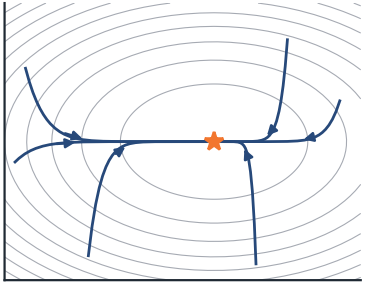
We consider the unconstrained minimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \qquad f \in C^1(\mathbb{R}^d).$$

The gradient $\nabla f(x)$ is the direction of maximal growth of f at x: moving along $-\nabla f(x)$ should lead to a minimizer $x^\star$.

Definition 3.1.1 (gradient flow)
The gradient flow of $f \in C^1(\mathbb{R}^d)$ from $x_0 \in \mathbb{R}^d$ is the ODE

$$\begin{cases} \dot{x} = -\nabla f(x) \\ x(0) = x_0. \end{cases}$$



Gradient flow trajectories: orthogonal to the level sets, into the minimizer.

Cauchy–Lipschitz gives *local* solutions. For global ones we assume:

Assumption 3.1.1
$f \in C^1(\mathbb{R}^d)$ and ∇f is Lipschitz continuous with constant $L > 0$: $\left\|\nabla f(x) - \nabla f(y)\right\|_2 \leq L \left\|x - y\right\|_2$ for all $x, y \in \mathbb{R}^d$.

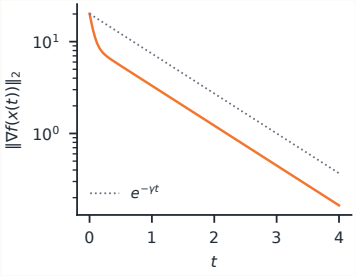
If $f \in C^2$: it suffices that $-L\,\mathrm{Id} \leq \mathrm{Hess}\, f(x) \leq L\,\mathrm{Id}$. Recall: $f \in C^2$ γ-strongly convex ($\mathrm{Hess}\, f \geq \gamma\,\mathrm{Id}$, $\gamma > 0$) $\implies$ unique minimizer $x^\star$.

Convergence of the gradient flow

Theorem 3.1.1
Let $f \in C^2(\mathbb{R}^d)$ be strongly convex and satisfy Assumption 3.1.1. Then, for any $x_0 \in \mathbb{R}^d$, the gradient flow converges to the unique minimizer $x^\star = \arg\min f$.

Proof sketch. Use $V(x) = \frac{1}{2}\left\|\nabla f(x)\right\|_2^2$ as a Lyapunov function:

- Along a solution, $\nabla V = \mathrm{Hess}\, f\, \nabla f$ and $\mathrm{Hess}\, f \geq \gamma\, \mathrm{Id}$ give $\frac{d}{dt}V(x(t)) \leq -2\gamma V(x(t))$, hence
$$\left\|\nabla f(x(t))\right\|_2 \leq \left\|\nabla f(x_0)\right\|_2 e^{-\gamma t} \longrightarrow 0 = \left\|\nabla f(x^\star)\right\|_2.$$
- The trajectory stays bounded: by co-coercivity (next slide),
$$\frac{d}{dt}\frac{1}{2}\left\|x(t) - x^\star\right\|_2^2 \leq -\frac{1}{L}\left\|\nabla f(x(t)) - \nabla f(x^\star)\right\|_2^2 \leq 0.$$
- Compactness: a subsequence $x(t_n) \to x_\infty$, and continuity of ∇f forces $\nabla f(x_\infty) = 0$, so $x_\infty = x^\star$ by uniqueness of the critical point. ■



The gradient along the flow of the quadratic of the previous slide: exponential decay.

Strong convexity turns the gradient flow into an exponentially stable dynamical system.

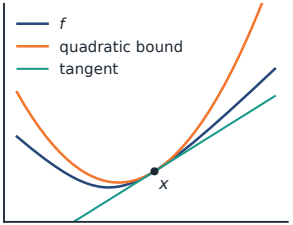
Co-coercivity

Theorem 3.1.2 (co-coercivity)
Let $f \in C^1(\mathbb{R}^d)$ be convex with ∇f L-Lipschitz. Then:

1. *Quadratic upper bound* (no convexity needed): $f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2, \quad \forall x, y \in \mathbb{R}^d.$
2. If $x^\star$ minimizes f: $\frac{1}{2L} \|\nabla f(y)\|_2^2 \le f(y) - f(x^\star) \le \frac{L}{2} \|y - x^\star\|_2^2, \qquad \forall y \in \mathbb{R}^d.$
3. *Co-coercivity*: $\|\nabla f(x) - \nabla f(y)\|_2^2 \le L \langle x - y, \nabla f(x) - \nabla f(y) \rangle, \qquad \forall x, y \in \mathbb{R}^d.$

Proof ideas.

- *1.* Integrate $f(y) = f(x) + \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle \, dt$, Lipschitz + Cauchy–Schwarz.
- *2.* Right side: 1. with $x = x^\star$. Left side: apply 1. at y and minimize over the second point: $f(x^\star) = \inf_z f(z) \le f(y) - \frac{1}{2L} \|\nabla f(y)\|_2^2$.
- *3.* Apply 2. to $f_x(z) = f(z) - \langle \nabla f(x), z \rangle$ (minimized at x) and to f_y, then sum the two inequalities. ■



Tangent below (convexity), quadratic of curvature L above (smoothness).

Gradient descent

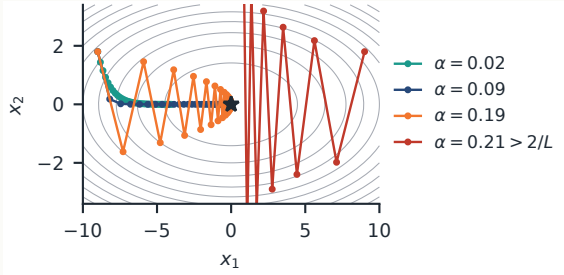
A natural time-discretization of the gradient flow (explicit Euler) is

$$\dot{x}(t) \sim \frac{x(t) - x(t+\alpha)}{\alpha}.$$

Definition 3.1.2 (gradient descent)
Starting at $x_0 \in \mathbb{R}^d$, with step sizes $(\alpha_n)_n \subset \mathbb{R}_+$:

$$x_{n+1} = x_n - \alpha_n \nabla f(x_n).$$

- α_n chosen adaptively or constant $\alpha_n = \alpha$.
- Too small $\rightsquigarrow$ slow; too large $\rightsquigarrow$ oscillations, divergence.
- Key quantity: the condition number L/γ. Elongated level sets force small steps and zig-zagging.



Ill-conditioned quadratic ($L = 10$): beyond $2/L$ the iterates diverge.

Convergence of gradient descent

Theorem 3.1.3

Let $f \in C^1(\mathbb{R}^d)$ be convex and coercive (i.e., $f(x) \to +\infty$ as $\|x\|_2 \to +\infty$), with ∇f L-Lipschitz and a unique minimizer $x^\star$. Assume

$$\sum_{n=0}^{+\infty} \alpha_n = +\infty \qquad \text{and} \qquad \alpha_n \leq \frac{1}{L}, \quad \forall n \in \mathbb{N}.$$

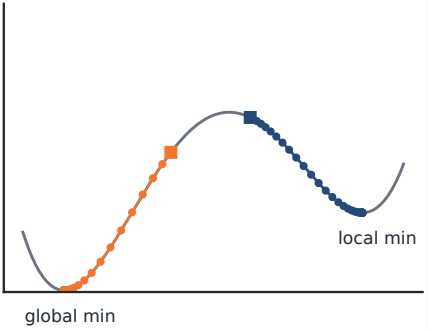
Then $x_n \to x^\star$ as $n \to +\infty$.

Proof sketch.

- *Descent step.* The quadratic upper bound with $y = x_{n+1} = x_n - \alpha_n \nabla f(x_n)$ and $\alpha_n \leq 1/L$ gives $f(x_{n+1}) \leq f(x_n) - \frac{\alpha_n}{2} \left\|\nabla f(x_n)\right\|_2^2$: $(f(x_n))_n$ decreases, and the iterates stay in a bounded sublevel set (coercivity).
- *Summability.* Telescoping: $\frac{1}{2} \sum_n \alpha_n \left\|\nabla f(x_n)\right\|_2^2 \leq f(x_0) - f(x^\star) < +\infty$.
- Since $\sum_n \alpha_n = +\infty$, this forces $\liminf_n \left\|\nabla f(x_n)\right\|_2 = 0$: a subsequence $x_{n_k} \to x_\infty$ with $\nabla f(x_\infty) = 0$, so $x_\infty = x^\star$ by convexity and uniqueness.
- $(f(x_n))_n$ is non-increasing with a subsequence reaching $f(x^\star)$, so $f(x_n) \to f(x^\star)$: every cluster point is a minimizer, hence $x_n \to x^\star$. ■

Remarks, and what survives without convexity

- Coercivity can be relaxed: it suffices that the sublevel set $\{f \le f(x_0)\}$ be bounded.
- Uniqueness of the minimizer can be removed: every cluster point of the iterates is then a minimizer.
- Without convexity: the same proof shows that every cluster point is a *critical point* of f. It need not be a minimizer (saddle points, local minima).



Two runs of gradient descent on a double well: the limit depends on the initialization.

This is the setting of deep learning: the loss is non-convex, and gradient methods only promise critical points. That they work so well in practice is (largely) a different story.

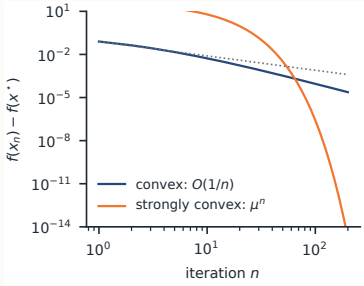
Rates of convergence

Theorem 3.1.4 (rates for constant step size $\alpha_n = \alpha$)
Under the assumptions of Theorem 3.1.3:

- f convex, $\alpha \le 1/L$: *sublinear* rate $\quad f(x_n) - f(x^\star) \le \dfrac{\|x_0 - x^\star\|_2^2}{2\alpha n}, \qquad \forall n \ge 1;$
- f γ-strongly convex, $\alpha < \gamma/L^2$: *exponential* rate, with $\mu = 1 - \alpha\gamma + \alpha^2 L^2 \in (0,1), \quad \|x_n - x^\star\|_2^2 \le \mu^n \|x_0 - x^\star\|_2^2.$

Proof ideas.

- *Convex:* combine the descent step with convexity at x_n, then telescope $f(x_{n+1}) - f(x^\star) \le \frac{1}{2\alpha}\big[\|x_n - x^\star\|_2^2 - \|x_{n+1} - x^\star\|_2^2\big]$.
- *Strongly convex:* expand $\|x_{n+1} - x^\star\|_2^2$, use $\langle \nabla f(x_n), x_n - x^\star\rangle \ge \frac{\gamma}{2}\|x_n - x^\star\|_2^2$ and the Lipschitz bound; recursion gives μ^n. ■



Optimality gap vs. iteration (log scale).

3.2

Numerical methods for (non)convex optimization

Stochastic gradient descent

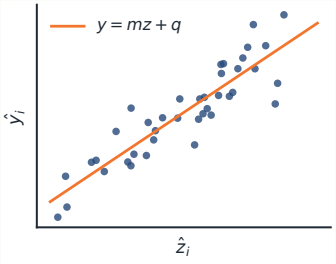
Why stochastic gradients? The machine learning setting

Gradient descent needs $\nabla f(x)$ at every step. In machine learning the objective is typically an expectation over the data:

$$f(x) = \mathbb{E}\left[f(x, \cdot)\right] = \int_\Omega f(x, \omega)\, d\mathbb{P}(\omega),$$

where $(\Omega, \mathcal{F}, \mathbb{P})$ is the space of training samples and $f(\cdot, \omega)$ is the loss on the sample ω.

Then the full gradient $\nabla f(x) = \int_\Omega \nabla f(x, \omega)\, d\mathbb{P}(\omega)$ requires a pass over the *whole* dataset: for large Ω, one gradient descent step is already too expensive.



Linear regression: one loss term per sample.

Example 3.2.1 (linear regression)

Samples $\omega = (\hat{z}, \hat{y})$, model $y = mz + q$, parameter $x = (m, q)$. Loss on one sample and over N samples:

$$\ell((m, q), (\hat{z}, \hat{y})) = (\hat{y} - m\hat{z} - q)^2, \qquad \ell((m, q)) = \frac{1}{N}\sum_{i=1}^{N} (\hat{y}_i - m\hat{z}_i - q)^2.$$

Here $\Omega = \{(\hat{z}_i, \hat{y}_i) \mid i \in [\![1, N]\!]\}$ with the uniform probability.

The algorithm

Idea: replace $\nabla f(x)$ by a sampled gradient, one sample per iteration.

Definition 3.2.1 (stochastic gradient descent) [Royer]
Let $f = \mathbb{E}[f(\cdot, \omega)]$ with $f(\cdot, \omega) \in C^1(\mathbb{R}^d)$ for all ω. Starting at x_0, with step sizes $(\alpha_n)_n$:

$$x_{n+1} = x_n - \alpha_n \nabla_x f(x_n, \omega_n),$$

where $(\omega_n)_n$ are i.i.d. random variables on Ω.

The iterates are now *random*: we control their behavior in expectation, under:

Assumption 3.2.1
At every iteration n:

1. *Unbiasedness:* $\mathbb{E}_{\omega_n}\left[\nabla f(x, \omega_n)\right] = \nabla f(x)$ for all $x \in \mathbb{R}^d$;
2. *Bounded variance:* there exists σ^2 such that $\mathbb{E}_{\omega_n}\left[\left\|\nabla f(x, \omega_n)\right\|_2^2\right] \leq \left\|\nabla f(x)\right\|_2^2 + \sigma^2$ for all x.

Example (uniform sampling): if $\Omega = [\![1, N]\!]$ and $\mathbb{P}(\omega_n = k) = 1/N$, then $\mathbb{E}_{\omega_n}\left[\nabla f(x, \omega_n)\right] = \frac{1}{N}\sum_{k=1}^{N} \nabla f(x, k) = \nabla f(x)$: unbiasedness always holds.

One step of SGD, in expectation

The results of this section are stated without proofs; see the notes and [Royer] for the details.

Proposition 3.2.1

Assume $f \in C^1(\mathbb{R}^d)$ with ∇f L-Lipschitz, and Assumption 3.2.1. Then

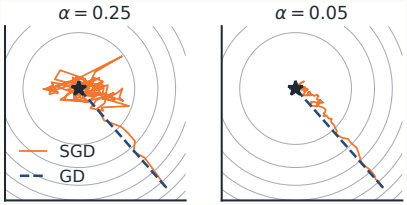
$$\mathbb{E}_{\omega_n}\left[f(x_{n+1})\right] \leq f(x_n) - \alpha_n\Big(1 - \frac{L\alpha_n}{2}\Big)\left\|\nabla f(x_n)\right\|_2^2 + \frac{L\alpha_n^2}{2}\sigma^2.$$

Compare with the deterministic descent step

$$f(x_{n+1}) \leq f(x_n) - \alpha_n\Big(1 - \frac{L\alpha_n}{2}\Big)\left\|\nabla f(x_n)\right\|_2^2$$

- same decrease term, driven by the true gradient;
- plus a noise term $\frac{L\alpha_n^2}{2}\sigma^2$, quadratic in the step size.

Small steps average the noise out ($\alpha_n^2 \ll \alpha_n$), but slow the descent down: the whole theory of SGD is the balance between these two effects.



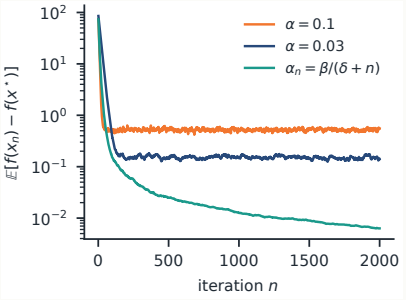
GD and SGD with the same step and the same samples. A large step descends fast but keeps jittering around $x^\star$, a small one is slower but settles closer.

Constant step size: fast, but only up to a noise floor

Theorem 3.2.1 (constant step size, without proof)
Let $f \in C^1(\mathbb{R}^d)$ be γ-strongly convex with ∇f L-Lipschitz, under Assumption 3.2.1. If $\alpha_n = \alpha$ with $0 < \alpha \le 1/L$, then

$$\mathbb{E}\left[f(x_n) - f(x^\star)\right] \le \underbrace{\frac{\alpha L \sigma^2}{2\gamma}}_{\text{noise floor}} + (1 - \alpha\gamma)^n \left[f(x_0) - f(x^\star) - \frac{\alpha L \sigma^2}{2\gamma}\right].$$

- Exponential decay *towards a ball* around the optimum, of radius proportional to $\alpha\sigma^2$.
- For any $\varepsilon > 0$ there are α and n_0 with $\mathbb{E}[f(x_n) - f(x^\star)] \le \varepsilon$ for $n \ge n_0$ — but in general $\lim_{n\to+\infty} \mathbb{E}\left[f(x_n) - f(x^\star)\right] \neq 0$.
- Practical reading: fast initial progress, then stagnation at the noise floor.



SGD on a strongly convex problem: constant steps stall at a floor proportional to α.

Decreasing step size: convergence, at a sublinear rate

Theorem 3.2.2 (variable step size, without proof)

Same assumptions, with step sizes

$$\alpha_n = \frac{\beta}{\delta + n}, \qquad \beta > \frac{1}{\gamma}, \qquad \alpha_0 = \frac{\beta}{\delta} < \frac{1}{L}.$$

Then

$$\mathbb{E}\left[f(x_n) - f(x^\star)\right] \leq \frac{v}{\delta + n}, \qquad \text{where} \qquad v = \max\left\{\delta\left[f(x_0) - f(x^\star)\right], \frac{\beta^2 L \sigma^2}{2(\beta\gamma - 1)}\right\}.$$

- Now $\mathbb{E}[f(x_n) - f(x^\star)] \to 0$: the decreasing steps shrink the noise floor to zero.
- The price: the rate is $O(1/n)$, sublinear — against the exponential rate of deterministic gradient descent on the same strongly convex problem.
- The step schedule $\alpha_n \sim \beta/n$ is exactly the classical Robbins–Monro regime: $\sum \alpha_n = \infty$, $\sum \alpha_n^2 < \infty$.

SGD trades per-iteration cost against rate: each step touches one sample instead of N, at the price of noise, which constant steps cannot eliminate and decreasing steps eliminate slowly.

3.3

Numerical methods for (non)convex optimization

Proximal methods

Composite problems: smooth + simple

The objective is often not smooth, but has structure:

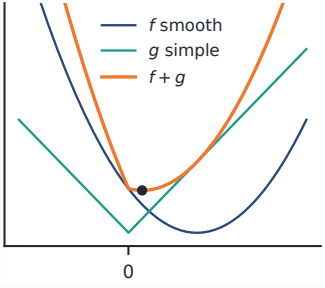
$$\min_{x\in\mathbb{R}^d} f(x) + g(x), \qquad f \text{ smooth } (C^1, \text{Lipschitz gradient}), \quad g \text{ non-smooth but "simple".}$$

Example

- *Constrained optimization:* with χ_K the 0-∞ characteristic function of a set K,

$$\min_{x\in K} f(x) = \min_{x\in\mathbb{R}^d} f(x) + \chi_K(x).$$

- *Lasso regression:* $\min_{x\in\mathbb{R}^d} \|Ax - b\|_2^2 + \lambda \|x\|_1$.



Smooth f, simple non-smooth g: the sum has a kink where g does.

Recall from Section 1.5 the proximal operator of a convex, lower semicontinuous (l.s.c.) function g:

$$P_g(y) = \arg\min_{x\in\mathbb{R}^d} \left[g(x) + \frac{1}{2} \|x - y\|_2^2 \right].$$

Function g "simple" means: P_g computable in closed form or cheaply.

Definition 3.3.1 (proximal gradient method)

For $f \in C^1(\mathbb{R}^d)$ and g convex, starting at x_0, with step sizes $(\alpha_n)_n$:

$$x_{n+1} = P_{\alpha_n g}\left[x_n - \alpha_n \nabla f(x_n)\right].$$

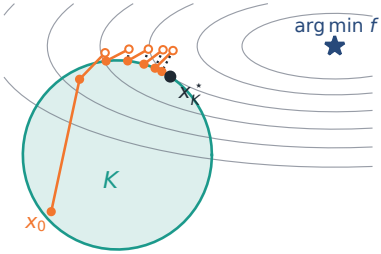
The iteration is a two step process:

1. Gradient step on the smooth part: $y_{n+1} = x_n - \alpha_n \nabla f(x_n)$;
2. Correction by the non-smooth part:

$$x_{n+1} = P_{\alpha_n g}(y_{n+1}) = \arg\min_{x \in \mathbb{R}^d} \left[g(x) + \frac{1}{2\alpha_n} \left\| x - y_{n+1} \right\|_2^2 \right],$$

a point *near* y_{n+1} that makes g small.

- If g is smooth: $x_{n+1} = x_n - \alpha_n\big(\nabla f(x_n) + \nabla g(x_{n+1})\big)$, implicit in g (not plain gradient descent).
- If $g = \chi_K$, K convex: P_{ag} is the projection onto K, and the method is the projected gradient descent. (*Exercise.*)



Projected gradient descent for $g = \chi_K$: the gradient step (solid) may leave K, the projection (dotted) brings it back. The iterates slide along ∂K to $x_K^\star$.

Convergence

Theorem 3.3.1
Let $g : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be proper, convex, l.s.c., and $f \in C^1(\mathbb{R}^d)$ convex with ∇f L-Lipschitz. Assume $f + g$ coercive with unique minimizer $x^\star$. Then the proximal gradient iterates with $\alpha_n = \alpha < 1/L$ converge to $x^\star$, with the same rates as in Theorem 3.1.4.

Proof sketch (convergence; see the notes for the rates).

- *Descent.* Comparing $x^+ = P_{\alpha g}(x - \alpha \nabla f(x))$ with the competitor $z = x$ in the definition of the proximal operator, and adding the quadratic upper bound for f:

$$F(x_{n+1}) \le F(x_n) - \frac{c}{2} \left\| x_{n+1} - x_n \right\|_2^2, \qquad F = f + g, \quad c = \alpha^{-1} - L > 0.$$

- *Telescoping* gives $\sum_n \left\| x_{n+1} - x_n \right\|_2^2 < +\infty$, so $\left\| x_{n+1} - x_n \right\|_2 \to 0$; coercivity bounds the iterates.
- *Identifying the limit.* The optimality condition of the prox step, in subgradient form:

$$-\nabla f(x_n) - \frac{1}{\alpha} \left(x_{n+1} - x_n \right) \in \partial g(x_{n+1}).$$

Along a convergent subsequence $x_{n_k} \to x_\infty$ (closedness of the graph of ∂g): $0 \in \nabla f(x_\infty) + \partial g(x_\infty) = \partial F(x_\infty)$, so $x_\infty = x^\star$, and the whole sequence converges. ∎

Example: Lasso and ISTA

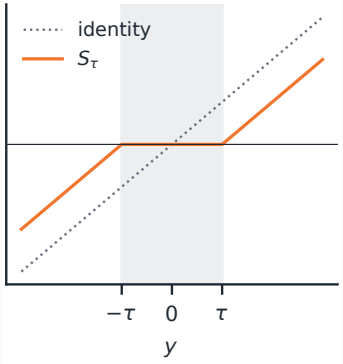
Example (regression with ℓ_1 regularization)
$\min_{x\in\mathbb{R}^d} \|Ax - b\|_2^2 + \lambda \|x\|_1$, with $A \in \mathbb{R}^{m\times d}$, $b \in \mathbb{R}^m$: the $\|x\|_1$ term promotes sparsity of $x^\star$. Composite splitting: $f(x) = \|Ax - b\|_2^2$, $g(x) = \lambda \|x\|_1$.

From Example 1.5.2, $P_{\alpha g}$ is the soft-thresholding operator $S_{\alpha\lambda}$, applied componentwise.

The proximal gradient method becomes the *iterative shrinkage-thresholding algorithm* (ISTA):

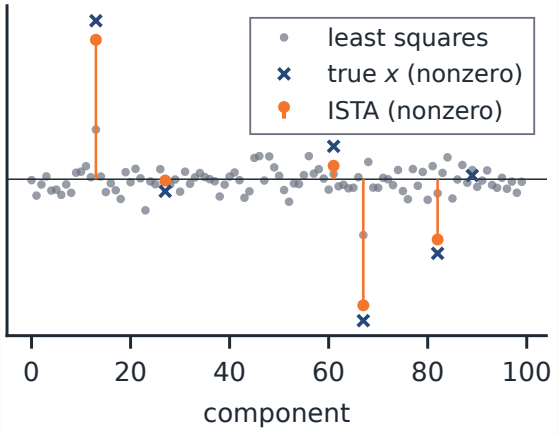
$$x_{n+1} = S_{\alpha\lambda}\left(x_n - 2\alpha A^\top (Ax_n - b)\right).$$

Each iteration: a gradient step on the least-squares term, then shrink every component towards 0 — small ones are set *exactly* to zero: this is where sparsity comes from.



The soft-thresholding function S_τ: an affine shrink outside $[-\tau, \tau]$, zero inside.

ISTA at work



A random instance with $A \in \mathbb{R}^{40\times100}$ and a 6-sparse true vector. ISTA is exactly zero off its five stems, and shrinks the entries it keeps.

- The system $Ax = b$ is underdetermined ($m < d$): least squares spreads the signal over every component.
- ISTA drives all the other components *exactly* to zero and recovers the support of the true vector — except its smallest component, which lies below the threshold: the price of the $\lambda \|x\|_1$ bias.
- This is the mechanism behind compressed sensing and sparse regression.

3.4

Numerical methods for (non)convex optimization

Dual methods

When the prox is hard: compose with a linear map

From Chapter 2: every convex problem has a dual problem, sometimes with better structure. We focus on

$$\min_{x \in \mathbb{R}^d} f(x) + g(Ax), \qquad A \in \mathbb{R}^{m \times d}, \quad f \text{ smooth convex}, \quad g \text{ non-smooth convex}.$$

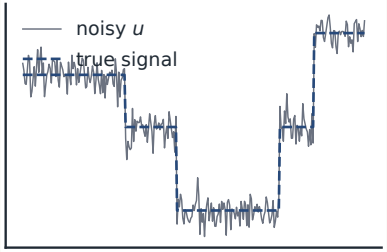
- If $P_{g \cdot A}$ is simple: use the proximal gradient method of the previous section.
- Often, however, P_g is simple but $P_{g \cdot A}$ is very hard to compute. The dual problem circumvents this.

Example (signal denoising by total variation)

Denoise a 1d signal $u \in \mathbb{R}^d$:

$$\min_{x \in \mathbb{R}^d} \|x - u\|_2^2 + \lambda \sum_{i=1}^{d-1} |x_{i+1} - x_i| .$$

This is $f(x) + g(Ax)$ with $f(x) = \|x - u\|_2^2$, $g = \|\cdot\|_1$, and A the difference operator $(Ax)_i = x_{i+1} - x_i$. The prox of g is soft-thresholding; the prox of $g \cdot A$ has no closed form.



The data: a piecewise-constant signal under noise. The ℓ_1 norm of the increments is the natural penalty.

The dual problem

Rewrite the problem by splitting the variable:

$$\min_{(x,y)\in\mathbb{R}^d\times\mathbb{R}^m} f(x) + g(y) \qquad \text{subject to} \qquad y = Ax.$$

Proposition 3.4.1

The dual problem is

$$\max_{\xi\in\mathbb{R}^m} \left[-f^*(-A^\top \xi) - g^*(\xi)\right],$$

where f^*, g^* are the convex conjugates of Section 1.4.

Proof. The Lagrange function is $L(x, y, \xi) = f(x) + g(y) + \langle \xi, Ax - y\rangle$, and the dual function

$$H(z) = \inf_{x,y} L(x, y, z) = \inf_x \{f(x) + \langle z, Ax\rangle\} + \inf_y \{g(y) - \langle z, y\rangle\} = -f^*(-A^\top z) - g^*(z). \qquad \blacksquare$$

Why this helps:

- $f^* \circ (-A^\top)$ inherits the good properties of f (smooth, Lipschitz gradient if f is strongly convex);
- P_{g^*} is as cheap as P_g, by Moreau's identity (Theorem 1.5.1): $P_{g^*}(y) = y - P_g(y)$.

Dual proximal gradient

Theorem 3.4.1 (dual proximal gradient)
Let g be proper, convex, l.s.c., and $f \in C^1(\mathbb{R}^d)$ be γ-strongly convex. The proximal gradient method applied to the dual problem reads

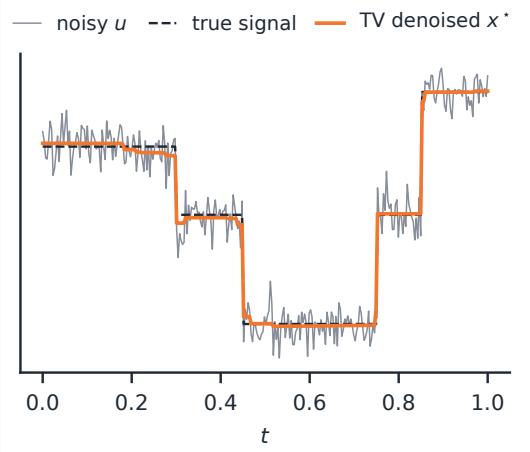
$$\begin{cases} x_{n+1} = \arg\min_{x \in \mathbb{R}^d} \left[f(x) + \langle z_n, Ax \rangle \right], \\ y_{n+1} = \arg\min_{y \in \mathbb{R}^m} \left[g(y) - \langle z_n, y \rangle + \frac{\alpha_n}{2} \left\| Ax_{n+1} - y \right\|_2^2 \right], \\ z_{n+1} = z_n + \alpha_n \left(Ax_{n+1} - y_{n+1} \right). \end{cases}$$

If the primal problem has a unique minimizer $x^\star$ and $\alpha_n = \alpha < \gamma / \|A\|^2$, then $x_n \to x^\star$.

Proof idea. Strong convexity of f makes $z \mapsto f^*(-A^\top z)$ C^1 with $\frac{\|A\|^2}{\gamma}$-Lipschitz gradient, and $\nabla f^*(v) = \arg\min_x [f(x) - \langle v, x \rangle]$. The prox step on g^* is converted into a prox step on g via Moreau's identity. ■

- Each line is easy: a smooth minimization in x, a prox of g in y, an explicit multiplier update.
- *Fixed points = KKT triples*: the third line gives primal feasibility $Ax^\star = y^\star$, the first two the stationarity $-A^\top z^\star \in \partial f(x^\star)$, $z^\star \in \partial g(y^\star)$; z_n approximates the optimal multiplier.
- For the TV denoising example, x_n and y_n have closed-form expressions.

Dual proximal gradient at work, and a summary of Chapter 3



Total-variation denoising by the dual proximal gradient method: the ℓ_1 penalty on the increments recovers a piecewise-constant signal.

Take-away of Chapter 3

- **Smooth:** gradient descent; $O(1/n)$ if convex, exponential if strongly convex.
- **Large sums/expectations:** SGD; noise floor for constant steps, $O(1/n)$ with $\alpha_n \sim \beta/n$.
- **Smooth + simple:** proximal gradient; same rates as gradient descent.
- **Smooth + simple $\circ$ linear:** dualize, then proximal gradient on the dual.

4

chapter

Controllability

- Control systems
- Controllability of linear autonomous systems
- Controllability of time-varying linear systems

4.1

Controllability

Control systems

Control systems

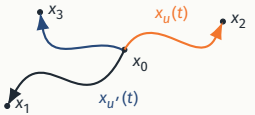
For this chapter we follow [Trélat].

Definition 4.1.1 (control system)
A control system in $\mathbb{R}^n$ is the differential equation

$$\dot{x} = f(t, x, u),$$

where

- the *state* is $x : \mathbb{R} \to \mathbb{R}^n$;
- $f : \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ is C^1 in (x, u) and locally integrable in t;
- the *control* $u : \mathbb{R} \to U$ is measurable, essentially bounded, with values in $U \subset \mathbb{R}^m$.



Each control u steers x_0 along its own trajectory $x_u(t)$.

The system is

- linear if $f(t, x, u) = A(t)x + B(t)u + r(t)$, with A, B, r of class L^∞ on compact intervals;
- autonomous if $f(t, x, u) = f(x, u)$; otherwise *time-varying*.

Optimization: choose a point $x \in \mathbb{R}^d$. Control: choose a *function* $u(\cdot)$.

Existence of trajectories, admissible controls

For a fixed control u, the vector field $t \mapsto f(t, x, u(t))$ is only measurable in time. The classical Cauchy–Lipschitz theory extends:

Theorem 4.1.1 (Carathéodory existence theorem)

Consider $\dot{x} = f(t, x)$, $x(0) = x_0$, with

- $f(t, \cdot)$ Lipschitz for every t, with locally integrable Lipschitz constant $L(t)$;
- $f(\cdot, x)$ measurable for every x;
- $\|f(t, x)\|_2 \le M$ on $(-r, r) \times B(0, r)$ for some $r, M > 0$.

Then there is a unique solution, maximally defined on an open interval $I \ni 0$.

Solutions may still blow up before a prescribed horizon T, whence:

Definition 4.1.2 (admissible controls)

$u \in L^\infty([0, T], U)$ is admissible on $[0, T]$ at x_0 if the trajectory x_u with $x_u(0) = x_0$ is well-defined on $[0, T]$. We set

$$\mathcal{U}_{x_0, T} = \{u \in L^\infty([0, T], U) \mid u \text{ admissible}\}.$$

Reachable sets and controllability

Definition 4.1.3 (controllability)

The end-point map and the reachable set from x_0 in time T are

$$\mathrm{End}_{x_0,T} : \mathcal{U}_{x_0,T} \to \mathbb{R}^n, \quad \mathrm{End}_{x_0,T}(u) = x_u(T), \qquad \mathrm{Reach}_{x_0,T} = \mathrm{End}_{x_0,T}(\mathcal{U}_{x_0,T}).$$

The system is

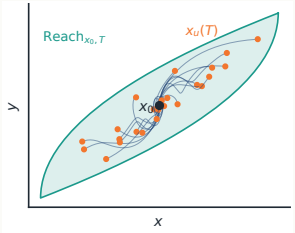
- *globally controllable* from x_0 in time T if $\mathrm{Reach}_{x_0,T} = \mathbb{R}^n$ ($\mathrm{End}_{x_0,T}$ surjective);
- *locally controllable* around $x_1 \in \mathrm{Reach}_{x_0,T}$ if $x_1 \in \operatorname{int} \mathrm{Reach}_{x_0,T}$.

Example 4.1.1 (trivial system)

$\dot{x} = u$, $u \in L^\infty([0,T], \mathbb{R}^d)$: then $\mathrm{End}_{x_0,T}(u) = x_0 + \int_0^T u(s)\,ds$, and the constant control $u \equiv (x_1 - x_0)/T$ steers x_0 to x_1. Globally controllable, from every point, in every time.

Example (constrained control)

$\dot{x} = u$ in $\mathbb{R}$ with $u \geq 0$: $\mathrm{Reach}_{x_0,T} = [x_0, +\infty)$ — never globally controllable, but locally controllable around every $x_1 > x_0$.



Double integrator $\dot{x} = y$, $\dot{y} = u$, $|u| \leq 1$: one trajectory per control, and $\mathrm{Reach}_{x_0,T}$ (shaded) collects the endpoints.

4.2

Controllability

Controllability of linear autonomous systems

Linear autonomous systems: two tools

We consider

$$\dot{x} = Ax + Bu, \qquad A \in \mathbb{R}^{n\times n},\ B \in \mathbb{R}^{n\times m}.$$

Linear systems do not blow up in finite time: $\mathcal{U}_{x_0,T} = L^\infty([0,T],U)$.

Variation of constants (Duhamel formula)

$$x_u(t) = e^{tA}x_0 + \int_0^t e^{(t-s)A}Bu(s)\,ds, \qquad \forall t \in [0,T],\ u \in L^\infty([0,T],U).$$

Theorem 4.2.1 (Cayley–Hamilton)

Let $\chi_A(z) = \det(z\,\mathrm{Id} - A) = z^n + a_1 z^{n-1} + \ldots + a_n$. Then $\chi_A(A) = 0$. That is,

$$A^n + a_1 A^{n-1} + \ldots + a_{n-1}A + a_n\,\mathrm{Id} = 0.$$

Consequence used below: every power A^j, $j \geq n$, is a linear combination of $\mathrm{Id}, A, \ldots, A^{n-1}$.

The Kalman rank condition

Definition 4.2.1 (Kalman rank condition)

The pair (A, B) satisfies the Kalman rank condition if the Kalman matrix

$$K = [B, AB, \dots, A^{n-1}B] \in \mathbb{R}^{n \times nm}$$

has maximal rank n.

Theorem 4.2.2 (Kalman)

Assume $U = \mathbb{R}^m$. The system $\dot{x} = Ax + Bu$ is globally controllable from $x_0 \in \mathbb{R}^n$ in time $T > 0$ if and only if (A, B) satisfies the Kalman rank condition.
In particular, controllability from one point in one time implies controllability from *every* point in *every* time.

- A purely algebraic test: no differential equation needs to be solved.
- Independent of x_0 and of T: a strong rigidity specific to the linear unconstrained case.
- The columns of K span the directions the control can excite, directly (B) or through the drift (AB, A^2B, ...); Cayley–Hamilton explains why powers beyond A^{n-1} bring nothing new.

Example: a system that fails the Kalman condition

Consider, in $\mathbb{R}^2$,

$$\begin{cases} \dot{x}_1 = u, \\ \dot{x}_2 = -x_2, \end{cases} \qquad A = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

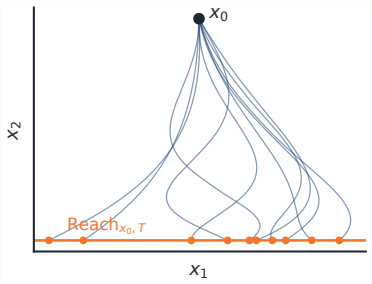
The Kalman matrix is

$$K = [B, AB] = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \operatorname{rank} K = 1 < 2.$$

The control only acts on x_1: whatever u, the second coordinate obeys $x_2(t) = e^{-t}x_2(0)$, and

$$\operatorname{Reach}_{x_0,T} = \mathbb{R} \times \{e^{-T}x_2(0)\},$$

a horizontal line — the uncontrollable direction of the Kalman decomposition (Proposition 4.2.1 below).



Trajectories for various controls from x_0: every endpoint lies on the same line.

Proof of Kalman's theorem

By Duhamel (and invertibility of e^{TA}), controllability $\iff$ surjectivity of

$$L : L^\infty([0,T],\mathbb{R}^m) \to \mathbb{R}^n, \qquad Lu = \int_0^T e^{-As} B u(s)\, ds.$$

- *Surjective* $\implies$ *Kalman.* If $\operatorname{rank} K < n$, take $p \neq 0$ with $p^\top A^i B = 0$ for $i \in [\![0, n-1]\!]$. By Cayley–Hamilton this holds for *all* $i \in \mathbb{N}$, so

$$p^\top (-As)^j B = 0 \quad \forall j \in \mathbb{N} \implies p^\top e^{-As} B = \sum_{j=0}^{+\infty} p^\top \frac{(-As)^j}{j!} B = 0 \implies p^\top Lu = 0 \quad \forall u.$$

  Hence, L is not surjective.

- *Kalman* $\implies$ *surjective.* If $p^\top Lu = 0$ for all u, test with $u = e_i \mathbf{1}_{[0,\tau]}$:

$$0 = p^\top \left[\frac{\mathrm{Id} - e^{-\tau A}}{A}\right] B e_i = \sum_{j=1}^{+\infty} \frac{(-1)^{j-1}\tau^j}{j!}\, p^\top A^{j-1} B e_i, \qquad \forall \tau \in [0,T].$$

  Analyticity in τ forces every coefficient to vanish: $p^\top A^{j-1} B e_i = 0$ for all j, i, i.e. $p^\top K = 0$ and $\operatorname{rank} K < n$. ■

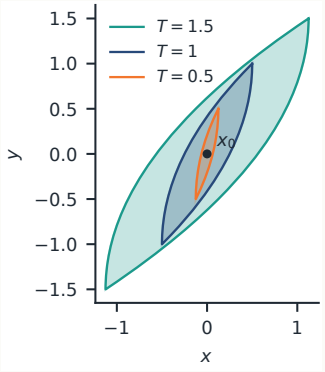
Both directions reduce to: the only p orthogonal to $\operatorname{Ran} L$ are those orthogonal to $\operatorname{Ran} K$.

Constrained controls: local controllability

Corollary 4.2.3 (control constraints)
Assume $0 \in \operatorname{int}(U)$ and the Kalman condition. Then, for every $x_0 \in \mathbb{R}^n$ and $T > 0$, the system is locally controllable around the drift endpoint: $e^{TA}x_0 \in \operatorname{int} \operatorname{Reach}_{x_0,T}$.

Proof.

- $\operatorname{End}_{x_0,T}(u) = e^{TA}x_0 + Lu$, with L linear and continuous: $\|Lu\|_2 \le T \max_{s\in[0,T]} \left\|e^{-As}B\right\| \|u\|_\infty$.
- Kalman $\implies$ L surjective onto $\mathbb{R}^n$ $\implies$ L is an open map.
- Hence the image of the small controls $V = \{\|u\|_\infty < \varepsilon\} \subset L^\infty([0,T], U)$ covers a neighborhood of $\operatorname{End}_{x_0,T}(0) = e^{TA}x_0$. ■



Reachable sets $\operatorname{Reach}_{0,T}$ of the double integrator $\dot{x} = y$, $\dot{y} = u$, $|u| \le 1$, for growing T.

The Hautus test

Theorem 4.2.4 (Hautus test)
The following are equivalent:

1. (A, B) satisfies the Kalman rank condition;
2. $\operatorname{rank}[\lambda \operatorname{Id} - A, B] = n$ for every $\lambda \in \mathbb{C}$;
3. $\operatorname{rank}[\lambda \operatorname{Id} - A, B] = n$ for every $\lambda \in \operatorname{spec} A$;
4. for every (possibly complex) eigenvector z of $A^\top$: $B^\top z \neq 0$;
5. there exists $c > 0$ such that, for all $z \in \mathbb{C}^n$ and $\lambda \in \mathbb{C}$,

$$\left\|(\lambda \operatorname{Id} - A^\top)z\right\|_2^2 + \left\|B^\top z\right\|_2^2 \geq c \left\|z\right\|_2^2 .$$

Proof ideas. 2 $\iff$ 3: $\lambda \operatorname{Id} - A$ is invertible off the spectrum. 3 $\iff$ 4: a rank defect of $[\lambda \operatorname{Id} - A, B]$ is a left eigenvector annihilated by $B^\top$. 2 $\iff$ 5: the left side of 5. is $z^* M_\lambda z$, $M_\lambda = (\bar{\lambda} \operatorname{Id} - A)(\lambda \operatorname{Id} - A^\top) + BB^\top$ Hermitian; take $c = \inf_\lambda \mu(\lambda) > 0$ with $\mu(\lambda)$ its smallest eigenvalue (μ continuous, $\to +\infty$ as $|\lambda| \to \infty$). 1 $\iff$ 4: if Kalman fails, $N = \{z \mid z^\top A^k B = 0 \ \forall k\} \neq \{0\}$ is $A^\top$-invariant, so it contains an eigenvector of $A^\top$, which 4. forbids. ■

Reading of 4.: the system is uncontrollable iff some mode of the adjoint dynamics is invisible to B.

Similar systems, and the controllable–uncontrollable splitting

Definition 4.2.2 (similar systems)
$\dot{x}_1 = A_1 x_1 + B_1 u$ and $\dot{x}_2 = A_2 x_2 + B_2 u$ are similar if there is $P \in \mathrm{GL}_n(\mathbb{R})$ with $A_2 = PA_1P^{-1}$ and $B_2 = PB_1$ (change of basis $x_2 = Px_1$).

The Kalman property is intrinsic: $K_2 = PK_1$, so similar systems have the same controllability. This leads to *normal forms*: coordinates in which the structure of the system becomes explicit.

Proposition 4.2.1 (Kalman decomposition)
If $\operatorname{rank} K = r < n$, then, in suitable coordinates $y = (y_1, y_2) \in \mathbb{R}^r \times \mathbb{R}^{n-r}$, the system reads

$$\begin{cases} \dot{y}_1 = A_1' y_1 + B_1' u + A_3' y_2 \\ \dot{y}_2 = A_2' y_2 \end{cases}$$

Namely, it splits in a controllable part y_1, and an uncontrollable part y_2 that evolves on its own.

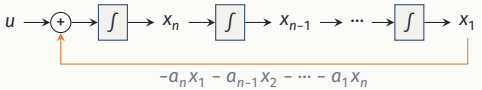
Proof idea. $F = \operatorname{Ran} K = \operatorname{Ran} B + \operatorname{Ran} AB + \dots + \operatorname{Ran} A^{n-1}B$ is A-invariant (Cayley–Hamilton) and contains $\operatorname{Ran} B$: choose a basis adapted to $\mathbb{R}^n = F \oplus G$; the matrices become block-triangular. ■

Brunovský normal form: single input

Theorem 4.2.5 (Brunovský, $m = 1$)
Let $B \in \mathbb{R}^{n\times 1}$ and (A, B) satisfy the Kalman condition, with $\chi_A(z) = z^n + a_1 z^{n-1} + \dots + a_n$. Then the system is similar to the chained form

$$\begin{cases} \dot{x}_1 = x_2, \\ \quad\vdots \\ \dot{x}_{n-1} = x_n, \\ \dot{x}_n = -a_n x_1 - a_{n-1} x_2 - \cdots - a_1 x_n + u. \end{cases}$$

Equivalently: every controllable single-input system is, in the right coordinates, the *n-th order scalar ODE* $y^{(n)} + a_1 y^{(n-1)} + \dots + a_n y = u$. That is, a chain of integrators with feedback:



Proof idea. Define $v_{n-k} = A^k B + a_1 A^{k-1} B + \dots + a_k B$, $k \in [\![0, n-1]\!]$ (so $v_n = B$). They form a basis (triangular change of basis from the full-rank K), and $Av_j = v_{j-1} - a_{n-j+1} v_n$ for $j \geq 2$, $Av_1 = (\chi_A(A) - a_n \mathrm{Id})B = -a_n v_n$ by Cayley–Hamilton: exactly the companion matrix. ∎

Brunovský normal form: general case

Theorem 4.2.6 (Brunovský, $m > 1$)
Let (A, B) satisfy the Kalman condition. Then there exist invertible $P \in \mathbb{R}^{n\times n}$ (change of state) and $R \in \mathbb{R}^{m\times m}$ (change of *input*) such that, with $x = P\tilde{x}$, $u = R\tilde{u}$, the system becomes $\dot{\tilde{x}} = A_c\tilde{x} + B_c\tilde{u}$ with the block-diagonal Brunovský form

$$A_c = \begin{bmatrix} A_1 & & \\ & \ddots & \\ & & A_m \end{bmatrix}, \qquad B_c = \begin{bmatrix} B_1 & & \\ & \ddots & \\ & & B_m \end{bmatrix},$$

where each (A_i, B_i) is a single-input Brunovský block of size r_i, and the *controllability indices* satisfy $r_1 + \cdots + r_m = n$.

- The system decomposes into m independent controllability chains (Jordan chains of the pair (A, B)).
- Note the extra ingredient compared to the single-input case: a linear change of the *input* is needed as well.

4.3

Controllability

Controllability of time-varying linear systems

Time-varying systems and the state-transition matrix

We now consider

$$\dot{x} = A(t)x + B(t)u, \qquad A : \mathbb{R} \to \mathbb{R}^{n\times n}, \quad B : \mathbb{R} \to \mathbb{R}^{n\times m}.$$

Definition 4.3.1 (state-transition matrix)

$R : \mathbb{R} \times \mathbb{R} \to \mathbb{R}^{n\times n}$ is the unique solution of

$$\frac{\partial}{\partial t} R(t, s) = A(t)R(t, s), \qquad R(s, s) = \mathrm{Id}\,.$$

Proposition 4.3.1

- Autonomous case $A(t) \equiv A$: $R(t, s) = e^{(t-s)A}$.
- *Semigroup property:* $R(t, s)R(s, \tau) = R(t, \tau)$; in particular $R(t, s)^{-1} = R(s, t)$.
- Solutions: $x(t) = R(t, 0)x_0 + \int_0^t R(t, s)B(s)u(s)\,ds$.

$R(t, s)$ replaces $e^{(t-s)A}$ everywhere; what it loses is the explicit exponential formula.

Why the naive Kalman condition fails, and the Gramian

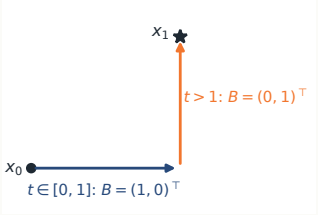
First guess: require the Kalman matrix $K(t)$ of $(A(t), B(t))$ to be full rank at each time. Too strong:

Example

In $\mathbb{R}^2$, $\dot{x} = B(t)u$ with

$$B(t) = \begin{cases} (1,0)^\top & t \in [0,1], \\ (0,1)^\top & t > 1. \end{cases}$$

At every fixed t, only one direction is available: $K(t)$ is never full rank. Yet $\mathrm{Reach}_{x_0,T} = \mathbb{R}^2$ for every $T > 1$: steer the first coordinate before $t = 1$, the second after.



Two phases, one direction each: together they reach the whole plane.

An instantaneous lack of controllability is harmless if the missing directions become available later. The right object *integrates* over time:

Definition 4.3.2 (controllability Gramian)

$$G_T := \int_0^T R(T,t)B(t)B(t)^\top R(T,t)^\top \, dt \in \mathbb{R}^{n\times n}.$$

The observability inequality

Proposition 4.3.2 (observability inequality)
G_T is symmetric and non-negative, and it is invertible if and only if there exists $C_T > 0$ with

$$\int_0^T \left\| B(t)^\top R(T,t)^\top \psi \right\|_2^2 \, dt \geq C_T \left\| \psi \right\|_2^2, \qquad \forall \psi \in \mathbb{R}^n.$$

Proof. $\psi^\top G_T \psi = \int_0^T \left\| B(t)^\top R(T,t)^\top \psi \right\|_2^2 \, dt \geq 0$: non-negativity, and invertibility of a non-negative symmetric matrix $\iff$ a uniform lower bound on its quadratic form. ■

Why "observability". Consider the adjoint system with measured output:

$$\dot{z} = -A(t)^\top z, \qquad z(T) = \psi, \qquad y(t) = B(t)^\top z(t).$$

Then $y(t) = B(t)^\top R(T,t)^\top \psi$ (backward in time), and the inequality reads

$$\int_0^T \left\| y(t) \right\|_2^2 \, dt \geq C_T \left\| z(T) \right\|_2^2.$$

That is, the energy of the output controls the size of the state: one can *reconstruct* $z(T)$ from the measurement y. In particular $y \equiv 0 \implies z(T) = 0$.

Controllability = observability of the adjoint

Theorem 4.3.1
Assume $U = \mathbb{R}^m$. The time-varying system is controllable from x_0 in time T if and only if G_T is invertible. In that case it is controllable from every initial point, and in every time $T' > T$.

Proof sketch. Write $\mathrm{End}_{x_0,T}(u) = x^\star + Lu$ with $x^\star = R(T,0)x_0$, $Lu = \int_0^T R(T,t)B(t)u(t)\,dt$.

- G_T *invertible* $\implies$ *controllable.* Look for $u(t) = B(t)^\top R(T,t)^\top \psi$: then $Lu = G_T\psi$, and
$$\psi = G_T^{-1}(x_1 - x^\star) \quad\Longrightarrow\quad \mathrm{End}_{x_0,T}(u) = x_1.$$
The steering control is *explicit*, given the Gramian.
- G_T *singular* $\implies$ *not controllable.* Take $\psi \neq 0$ in the kernel: the observability integral vanishes, so $B(t)^\top R(T,t)^\top \psi = 0$ a.e., hence $\psi^\top Lu = 0$ for every u and
$$\mathrm{Ran}\,\mathrm{End}_{x_0,T} \subset \{z \mid \psi^\top (z - x^\star) = 0\},$$
a proper affine subspace. ∎

Corollary 4.3.2 (control constraints)
If $0 \in \mathrm{int}(U)$ and G_T is invertible, the system is locally controllable from any x_0 around $R(T,0)x_0$.

A differential Kalman condition for time-varying systems

Theorem 4.3.3 (time-varying Kalman theorem)
Let $U = \mathbb{R}^m$ and $A(\cdot), B(\cdot)$ be C^∞. Define

$$B_0(t) = B(t), \qquad B_{j+1}(t) = A(t)B_j(t) - \frac{d}{dt}B_j(t), \qquad j \in \mathbb{N}.$$

If there exists $t_0 \in [0, T]$ such that

$$\operatorname{span} \bigcup_{j \in \mathbb{N}} \operatorname{Ran} B_j(t_0) = \mathbb{R}^n,$$

then the system is controllable in time T. If moreover $A(\cdot), B(\cdot)$ are analytic, this condition is equivalent to controllability, and independent of t_0.

- For constant A, B: $B_j = A^j B$, and one recovers exactly the Kalman rank condition.
- The correction $-\frac{d}{dt}B_j$ accounts for the motion of the directions $B_j(t)$ themselves: new directions can be generated either by the drift $A(t)$ or by the time variation of B.

Chapter 4 in one line: controllability of linear systems is a rank condition — algebraic (Kalman, Hautus) when autonomous, integrated (Gramian) or differential when time-varying.

5

chapter

Optimal control theory for linear systems

- Linear-quadratic optimal control problems
- The PMP as a Karush-Kuhn-Tucker condition
- Linear-quadratic problems with unconstrained terminal state

5.1

Optimal control theory for linear systems

Linear-quadratic optimal control problems

From controllability to optimality

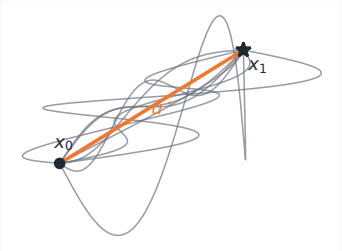
If the system is controllable, *many* controls steer x_0 to x_1 in time T. Which one is best? Fix a cost functional $J : \mathcal{U}_{x_0,T} \to \mathbb{R}$ and minimize it among the steering controls.

Definition (optimal control problem)

Given $x_0 \in \mathbb{R}^n$, $T > 0$, $x_1 \in \mathrm{Reach}_{x_0,T}$:

$$\begin{cases} \dot{x} = f(t, x, u), \\ x(0) = x_0, \quad x(T) = x_1, \\ J(u) \to \min. \end{cases}$$

(Variants: free endpoints, endpoint in a target set, free final time, ...)



Many trajectories from x_0 to x_1; the cost selects one.

With the admissible set $\Omega_{x_0,x_1,T} = \{u \in \mathcal{U}_{x_0,T} \mid x_u(T) = x_1\}$, this is an *infinite-dimensional constrained optimization problem*: $\min_{u \in \Omega_{x_0,x_1,T}} J(u)$.

Part I meets Part II: convexity, duality and KKT, applied to a constraint which is a differential equation.

Linear-quadratic problems

Setting for this chapter [Jurdjevic]: linear autonomous system, quadratic cost.

$$\text{(LQ)} \qquad \begin{cases} \dot{x} = Ax + Bu, \\ x(0) = x_0, \quad x(T) = x_1, \\ J(u) = \displaystyle\int_0^T c(x_u(t), u(t))\, dt \to \min, \end{cases}$$

with a quadratic running cost

$$c(x,u) = \frac{1}{2} u^\top P u + u^\top Q x + \frac{1}{2} x^\top R x, \qquad P \in \mathrm{Sym}(\mathbb{R}^m),\ Q \in \mathbb{R}^{m\times n},\ R \in \mathrm{Sym}(\mathbb{R}^n).$$

Theorem 5.1.1 (existence)
An optimal solution of (LQ) exists if and only if

$$J(u) \geq 0 \qquad \forall u \in \Omega_{0,0,T}.$$

If moreover $J(u) > \alpha \|u\|_2^2$ on $\Omega_{0,0,T}$ for some $\alpha > 0$, the optimal solution is unique.

Note where the condition lives: on the controls steering 0 to 0 — the *variations* available at fixed endpoints.

Existence: the structure of the proof

Key remark (linearity). For $u \in \Omega_{x_0,x_1,T}$ and $\tilde{u} \in \Omega_{0,0,T}$:

$$u_\lambda = u + \lambda\tilde{u} \in \Omega_{x_0,x_1,T} \quad \forall \lambda \in \mathbb{R}, \qquad \text{and} \qquad x_\lambda = x + \lambda\tilde{x}.$$

Expanding the quadratic cost:

$$J(u_\lambda) = J(u) + \lambda \int_0^T \left[\tilde{u}^\top P u + \tilde{u}^\top Q x + u^\top Q \tilde{x} + \tilde{x}^\top R x\right] dt + \lambda^2 J(\tilde{u}).$$

- *Necessity.* If u is optimal, $\lambda \mapsto J(u_\lambda)$ has a minimum at $\lambda = 0$: the quadratic in λ must have non-negative leading coefficient, i.e. $J(\tilde{u}) \geq 0$ on $\Omega_{0,0,T}$.
- *Sufficiency (sketch).* Fix $u_0 \in \Omega_{x_0,x_1,T}$; minimizing J over $\Omega_{x_0,x_1,T}$ is minimizing $\phi(\tilde{u}) = J(\tilde{u}) + L\tilde{u}$ (quadratic + linear) over $\tilde{u} \in \Omega_{0,0,T}$. Work in the Hilbert space $X \subset L^2$ of controls steering 0 to 0; on the orthogonal complement of the null space $N = \{J = 0\}$ (where L also vanishes, or ϕ would be unbounded below), J is coercive, so ϕ is continuous, coercive, strictly convex: a unique minimizer exists there, which minimizes ϕ on all of X.
- *Uniqueness.* $J(u) \geq \alpha \|u\|_2^2$ makes J strictly convex. ■

Same mechanism as in Chapter 1: existence of minimizers = coercivity + convexity (here of a quadratic functional on a Hilbert space of controls).

Standing assumptions

Assumption

- (A, B) satisfies the Kalman rank condition;
- there exists $T_{\max} > 0$ such that $J \geq 0$ on $\Omega_{0,0,T_{\max}}$;
- *dissipativity* holds: for every $T \in (0, T_{\max}]$ there exists $u \in \Omega_{0,0,T}$ with $J(u) > 0$.

The first gives controllability (the problem makes sense), the second existence, the third uniqueness.

Proposition 5.1.1
For $u \in \Omega_{x_0,x_1,T}$ and $\varepsilon > 0$ there is $u_\varepsilon \in \Omega_{x_0,x_1,T}$ with $J(u_\varepsilon) = J(u) + \varepsilon$.

Proof: $\lambda \mapsto J(u + \lambda\bar{u})$ is quadratic with positive leading coefficient $J(\bar{u}) > 0$; intermediate value theorem. ■

Proposition 5.1.2
Necessarily $P \in \mathrm{Sym}_{\geq 0}(\mathbb{R}^m)$.

Proof idea: if $P < 0$ on a subspace E_2, highly oscillating controls with values in E_2 have $\|x\|_\infty \leq \varepsilon \|u\|_{L^2}$ (their effect on the state averages out), so $J(u) \approx \frac{1}{2} \int u^\top P u \, dt$ can be made arbitrarily negative on $\Omega_{0,0,T}$. ■

Conversely, $P > 0$ guarantees $J \geq 0$ on $\Omega_{0,0,T}$ for T small (the u-term dominates for short times).

The extended system: optimality is a boundary property

Append the running cost to the state: on $(x, \bar{x}) \in \mathbb{R}^n \times \mathbb{R}$, consider

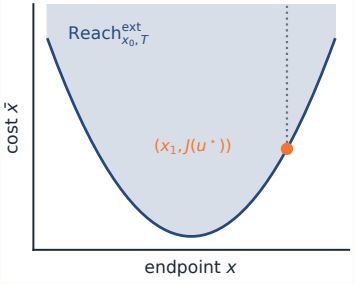
$$\begin{cases} \dot{x} = Ax + Bu \\ \dot{\bar{x}} = c(x, u) \end{cases} \quad \text{with initial condition } (x(0), \bar{x}(0)) = (x_0, 0).$$

with extended reachable set $\mathrm{Reach}^{\mathrm{ext}}_{x_0,T}$: every point of it is a pair (endpoint, cost) = $(x_u(T), J(u))$.

Theorem 5.1.2 (optimality and the extended system)
$u^\star \in \Omega_{x_0,x_1,T}$ is optimal $\iff$ $(x_1, J(u^\star)) \in \partial\, \mathrm{Reach}^{\mathrm{ext}}_{x_0,T}$.

Proof sketch.

- ($\Rightarrow$) The optimal pair is the *lowest* point of $\mathrm{Reach}^{\mathrm{ext}}_{x_0,T}$ on the vertical line $\{x_1\} \times \mathbb{R}$, hence on the boundary.
- ($\Leftarrow$) If $\bar{u}$ is *not* optimal, a cheaper control v exists; by openness of the end-point map (controllability) and Proposition 5.1.1 (costs can be raised at will), a full neighborhood of $(x_1, J(\bar{u}))$ lies in $\mathrm{Reach}^{\mathrm{ext}}_{x_0,T}$ — contradicting the boundary property. ∎



$\dot{x} = u$, $J = \frac{1}{2}\int u^2$: the extended reachable set is the region above the minimal-cost curve, and optimal points are its lower boundary.

The Pontryagin Maximum Principle

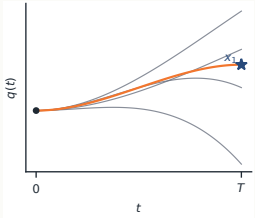
Theorem 5.1.3 (PMP for LQ problems)
Let $u^\star \in \Omega_{x_0,x_1,T}$ be optimal, with trajectory $x^\star$. Then there exists a nontrivial Lipschitz function $p : [0,T] \to \mathbb{R}^n$ (the costate, or adjoint state) such that

$$\begin{cases} \dot{x}^\star = Ax^\star + Bu^\star, \\ \dot{p} = -A^\top p + Rx^\star + Q^\top u^\star, \end{cases} \qquad \text{and} \qquad u^\star(t) = \arg\max_{v \in \mathbb{R}^m} H(x^\star(t), p(t), v),$$

where the Hamiltonian is

$$H(x,p,u) = p^\top (Ax + Bu) - c(x,u).$$

- Two coupled linear ODEs (state forward from x_0, costate with no prescribed initial value) + a pointwise maximization defining $u^\star$ from $(x^\star, p)$.
- The maximization is meaningful: $H(x,p,\cdot)$ is *concave*, since $P \geq 0$ (Proposition 5.1.2); for $P > 0$ it gives $u^\star = P^{-1}(B^\top p - Qx^\star)$.
- $p(0)$ is unknown: it is found by *shooting* (choose $p(0) = \lambda$, solve, adjust λ until $x^\star(T) = x_1$).



Shooting for $\ddot{q} = u$: each guess of $p(0)$ gives an extremal; one hits the target.

Proof of the PMP

$(x_1, J(u^\star)) \in \partial\,\mathrm{Reach}^{\mathrm{ext}}_{x_0,T}$. Consider variations $u_\lambda = u^\star + \lambda\tilde{u}$, so $x_\lambda = x^\star + \lambda\tilde{x}$, and the map $F(u) = (x_u(T), J(u))$ with directional derivative

$$L(\tilde{u}) := DF_{u^\star}(\tilde{u}) = \Big(\tilde{x}(T),\ \int_0^T \big[\partial_u c(x^\star, u^\star)\,\tilde{u} + \partial_x c(x^\star, u^\star)\,\tilde{x}\big]\,dt\Big).$$

- Ran L *is a graph.* $\mathrm{Ran}\,L \subset \mathbb{R}^{n+1}$ is a subspace; controllability makes its projection on the first n coordinates surjective, so $\dim \mathrm{Ran}\,L \geq n$. If $\mathrm{Ran}\,L$ were all of $\mathbb{R}^{n+1}$, some variation v would give $L(v) = (0, -\alpha)$, i.e. *same endpoint, strictly smaller cost* to first order — contradicting optimality. Hence
$$L(\tilde{u}) = \big(\tilde{x}(T),\ p_T(\tilde{x}(T))\big) \qquad \text{for a linear form } p_T : \mathbb{R}^n \to \mathbb{R}.$$
- *From p_T to the costate.* Solve the adjoint equation $\dot{p} = -A^\top p + \partial_x c(x^\star, u^\star)$ backwards from $p(T) = p_T$. Then, integrating $\frac{d}{dt}(p^\top \tilde{x})$ and comparing with $L(\tilde{u})$:
$$\int_0^T \big[\partial_u c(x^\star, u^\star) - B^\top p\big]\,\tilde{u}\;dt = 0 \quad \forall \tilde{u} \quad\Longrightarrow\quad \partial_u c(x^\star, u^\star) = B^\top p \quad \text{a.e.}$$
- This is $\partial_u H = 0$; concavity of H in u upgrades it to the maximization condition. ■

5.2

Optimal control theory for linear systems

The PMP as a Karush-Kuhn-Tucker condition

A change of viewpoint

The PMP was proved by *geometric* means. We now re-derive it from Chapter 2: it is the KKT theorem for a problem whose constraint is a differential equation. For simplicity, $Q = 0$:

$$c(x,u) = \frac{1}{2}\left(x^\top R x + u^\top P u\right).$$

The move: stop viewing x as a function of u. Treat the *pair* (x,u) as the unknown, and the dynamics as a *constraint*:

$$\min_{(x,u)\in \mathrm{Lip}([0,T],\mathbb{R}^n)\times L^\infty([0,T],\mathbb{R}^m)} J(x,u) \quad \text{s.t.} \quad \dot{x} - Ax - Bu = 0, \quad x(0) = x_0, \quad x(T) = x_1.$$

All constraints are *equalities* — the analogue of problem (OP) of Chapter 2 with no inequality constraints ($M = 0$) — except that the constraint $\dot{x} - Ax - Bu = 0$ is one equation per time t. Its multiplier is therefore a *function* of time: the costate.

The Lagrange function

$$\mathcal{L}(x,u,p) = \frac{1}{2}\int_0^T \left(x^\top R x + u^\top P u\right) dt + \int_0^T p^\top (\dot{x} - Ax - Bu)\, dt, \qquad p \in \mathrm{Lip}([0,T],\mathbb{R}^n).$$

The pairing $\int_0^T p^\top (\dot{x} - Ax - Bu)\, dt$ plays the role of $\langle \xi, Ax - y\rangle$.

The dictionary

Finite-dimensional problem (Ch. 2)	Optimal control problem (Ch. 5)
variable $x \in \mathbb{R}^d$	pair (x, u), trajectory and control
equality constraint $Ax - y = 0$	$\dot{x} - Ax - Bu = 0$, one equation per $t \in [0, T]$
multiplier $\xi \in \mathbb{R}^m$	costate $p \in \mathrm{Lip}([0, T], \mathbb{R}^n)$
pairing $\langle \xi, Ax - y \rangle$	$\int_0^T p^\top (\dot{x} - Ax - Bu)\, dt$
$L = F_0 + \langle \xi, Ax - y \rangle$	$\mathcal{L} = \int_0^T c\, dt + \int_0^T p^\top (\dot{x} - Ax - Bu)\, dt$
stationarity $\nabla_x L = 0$	adjoint equation and maximization condition
stationarity $\nabla_\xi L = 0$	the state equation itself
endpoint constraints	boundary terms: transversality conditions
constraint qualification	controllability of (A, B)
inequality constraints, $v \geq 0$	control constraints $u(t) \in U$

The PMP is the KKT stationarity condition, and the adjoint state is the Lagrange multiplier of the dynamics.

Stationarity

Integrate $\int_0^T p^\top \dot{x}\, dt$ by parts, so that $\mathcal{L}$ can be differentiated in x directly:

$$\mathcal{L}(x,u,p) = \frac{1}{2}\int_0^T \left(x^\top R x + u^\top P u\right) dt - \int_0^T \left(\dot{p}^\top x + p^\top (Ax+Bu)\right) dt + \underbrace{p(T)^\top x(T) - p(0)^\top x(0)}_{\text{boundary terms}}.$$

Stationarity $\partial_{(x,u)}\mathcal{L}(x^\star,u^\star,p) = 0$ (localizing variations in time gives pointwise conditions):

- *Variation in u:* $0 = Pu^\star - B^\top p \implies u^\star = P^{-1}B^\top p$ (the optimal control);
- *Variation in x:* $0 = Rx^\star - \dot{p} - A^\top p \implies \dot{p} = -A^\top p + Rx^\star$ (the adjoint equation);
- *Variation in p:* returns $\dot{x}^\star = Ax^\star + Bu^\star$ (as $\nabla_\xi L = 0$ returns the constraint).

These are the PMP conditions for $Q = 0$. On the Hamiltonian $H(x,p,u) = p^\top(Ax+Bu) - c(x,u)$:

$$\frac{\partial H}{\partial u} = B^\top p - Pu, \qquad \frac{\partial H}{\partial x} = A^\top p - Rx,$$

so the three conditions read $\partial_u H = 0$, $\dot{p} = -\partial_x H$, $\dot{x} = \partial_p H$: the Hamiltonian form of the PMP. (H is, up to sign, the integrand of $\mathcal{L}$.)

Transversality conditions

The boundary terms $p(T)^\top x(T) - p(0)^\top x(0)$ decide the boundary conditions on the costate:

A prescribed endpoint leaves the corresponding value of p free; a free endpoint prescribes it.

- A variation δx contributes $p(T)^\top \delta x(T) - p(0)^\top \delta x(0)$ to $\delta \mathcal{L}$.
- *Both endpoints fixed* (our problem): $\delta x(0) = \delta x(T) = 0$, the contribution vanishes, and stationarity says nothing about $p(0), p(T)$ — consistent with the shooting indeterminacy in Theorem 5.1.3.
- *Free terminal state* with final cost $\frac{1}{2} x(T)^\top W x(T)$ (next section): $\delta x(T)$ is now arbitrary, and

$$\big(W x^\star(T) + p(T)\big)^\top \delta x(T) = 0 \quad \forall \delta x(T) \qquad \Longrightarrow \qquad p(T) = -W x^\star(T).$$

This is exactly the terminal condition of Theorem 5.3.1 below: not an extra axiom, but the same stationarity principle. Conditions obtained this way are called transversality conditions.

Constraint qualification, and why a *maximum* principle

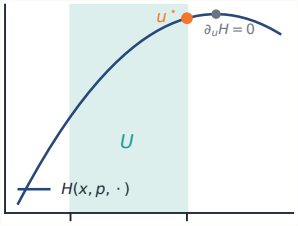
Constraint qualification. The general multiplier rule for min J s.t. $G = 0$ is the Fritz John form $p_0\, DJ_{u^\star} + \xi \circ DG_{u^\star} = 0$, $(p_0, \xi) \neq 0$, $p_0 \geq 0$. KKT is the normalization $p_0 = 1$, licit exactly when $DG_{u^\star}$ is surjective. Here the constraint map is the end-point map, whose differential is $\mathrm{End}_{0,T}$ itself: its surjectivity *is* controllability. When it fails, one may be forced to $p_0 = 0$: the *abnormal* case, where the multiplier carries no information about the cost.

The constraint qualification of the PMP is controllability of (A, B).

Why a maximum principle. Unconstrained controls: stationarity $\partial_u H = 0$ and concavity already give maximization. With constraints $u(t) \in U$ (saturating actuators), variations are one-sided, and stationarity becomes a *normal cone* inclusion (Section 2.4): a.e. in t,

$$\frac{\partial H}{\partial u}\big(x^\star, p, u^\star\big) \in \mathcal{N}_U\big(u^\star\big) \overset{\text{Prop. 2.4.1}}{\iff} u^\star \in \arg\max_{v \in U} H\big(x^\star, p, v\big).$$

For nonlinear systems this is *genuinely stronger* than stationarity (needle variations; [Agrachev–Sachkov, Trélat]).



Saturation: the critical point of H lies outside U; the maximizer sits on the boundary.

Optimal control theory for linear systems

5.3 Linear-quadratic problems with unconstrained terminal state

Releasing the terminal state

Shooting for $p(0)$ is unpleasant. Alternative: drop the constraint $x(T) = x_1$ and penalize the final state instead — compare with the penalty methods of Section 2.3:

$$\begin{cases} \dot{x} = Ax + Bu, & x(0) = x_0, \\ J(u) = \displaystyle\int_0^T c(x,u)\,dt + \frac{1}{2}x(T)^\top W x(T) \to \min, \end{cases} \qquad W \in \mathrm{Sym}_{\geq 0}(\mathbb{R}^n).$$

Assumption

(A, B) Kalman; $c(x,u) = \frac{1}{2}u^\top P u + \frac{1}{2}x^\top R x$ with $P \in \mathrm{Sym}_{>0}(\mathbb{R}^m)$, $R \in \mathrm{Sym}_{\geq 0}(\mathbb{R}^n)$.

Theorem 5.3.1 (PMP with terminal cost)

There is a unique optimal control $u^\star$, and a nontrivial Lipschitz costate $p^\star$ with

$$\begin{cases} \dot{x}^\star = Ax^\star + Bu^\star, & x^\star(0) = x_0, \\ \dot{p}^\star = -A^\top p^\star + Rx^\star, & p^\star(T) = -Wx^\star(T), \end{cases} \qquad u^\star = P^{-1}B^\top p^\star.$$

The transversality condition $p^\star(T) = -Wx^\star(T)$ replaces the lost boundary condition: the two-point boundary value problem is now well-posed — and, as we show next, solvable by a *matrix* ODE.

The value function

Definition 5.3.1 (value function)

$$V_T(x_0) = \inf\{J(u) \mid x_u(0) = x_0\}:$$

the optimal cost as a function of the initial state.

Proposition 5.3.1

For every $T > 0$ there exists $c > 0$ such that

$$V_s(x_0) \le c \left\|x_0\right\|_2^2, \qquad \forall x_0 \in \mathbb{R}^n,\ s \in [0,T].$$

Proof. Test with $u \equiv 0$: then $x(t) = e^{tA}x_0$ and

$$V_s(x_0) \le J(0) = \frac{1}{2}\left|\int_0^s x^\top R x\,dt + x(s)^\top W x(s)\right| \le \frac{1}{2}\left[\int_0^s \left\|e^{tA}\right\|^2 \|R\|\,dt + \|W\|\left\|e^{sA}\right\|^2\right]\left\|x_0\right\|_2^2. \qquad \blacksquare$$

This quadratic bound is the *a priori estimate* that will keep the Riccati equation from blowing up.

The Riccati equation and the optimal feedback law

The PMP gives $u^\star$ through $p^\star$, which is only known implicitly. For LQ problems one can do better: express $p^\star$ as a linear function of the state, $p^\star = E(t)x^\star$.

Theorem 5.3.2 (Riccati)

The unique optimal control is the feedback law

$$u^\star(t) = P^{-1}B^\top E(t)\, x^\star(t),$$

where $E : [0,T] \to \mathrm{Sym}(\mathbb{R}^n)$ is the unique solution of the matrix Riccati equation

$$\dot{E} = R - A^\top E - EA - EBP^{-1}B^\top E, \qquad E(T) = -W.$$

Moreover, the value function is the quadratic form $V_T(x_0) = -\frac{1}{2}x_0^\top E(0)x_0$.

- E is integrated *backwards* from $-W$, once, offline; the control is then a time-varying linear feedback $u^\star = K(t)x^\star$ — no shooting, and robust to perturbations of the state.
- Symmetry is preserved: $E(T) = -W$ is symmetric, and the right-hand side is symmetric in E.

Riccati: proof

Step 1: the feedback satisfies the PMP. Let E solve the Riccati equation (existence on $[0, T]$: next slide), let x solve $\dot{x} = Ax + BP^{-1}B^\top Ex$, and set $p = Ex$. Then

$$\dot{p} = \dot{E}x + E\dot{x} = \left(R - A^\top E - EA - EBP^{-1}B^\top E\right)x + E\left(Ax + BP^{-1}B^\top Ex\right) = -A^\top p + Rx,$$

and $p(T) = E(T)x(T) = -Wx(T)$: the triple (u, x, p) satisfies all PMP conditions. These are also *sufficient* here (convex problem), so by uniqueness $u = u^\star$, $x = x^\star$, $p = p^\star$.

Step 2: the value. Along the optimal trajectory, using $p = Ex$ and $p^\top Bu = u^\top Pu$:

$$\frac{1}{2}\frac{d}{dt}\, x^\top Ex = \frac{1}{2}\left(\dot{p}^\top x + p^\top \dot{x}\right) = \frac{1}{2}\left(x^\top Rx + u^\top Pu\right) = c(x, u).$$

Integrating and using $E(T) = -W$:

$$V_T(x_0) = \int_0^T c(x^\star, u^\star)\, dt + \frac{1}{2}x(T)^\top Wx(T) = -\frac{1}{2}x_0^\top E(0)x_0. \qquad \blacksquare$$

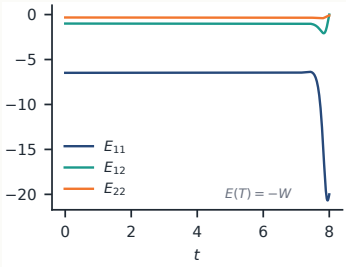
No blow-up: global existence of the Riccati solution

The Riccati equation is *quadratic* in E: solutions can blow up in finite time. We must rule this out on $[0, T]$.

Lemma 5.3.3
The solution of the Riccati equation with $E(T) = -W$ is defined on all of $[0, T]$.

Proof sketch. Suppose $\|E(t)\| \to +\infty$ as $t \downarrow t_\star$. By symmetry of E, the quadratic form $\dfrac{|x_0^\top E(t_0) x_0|}{\|x_0\|_2^2}$ then becomes arbitrarily large for suitable t_0, x_0. But solving the problem on $[t_0, T]$ and repeating the value-function computation gives $V_{T-t_0}(x_0) = -\frac{1}{2} x_0^\top E(t_0) x_0$, and $0 \le V_{T-t_0}(x_0) \le c \|x_0\|_2^2$ *uniformly* in $t_0 \in [0, T]$ (Proposition 5.3.1, plus $R, W \succeq 0$, $P > 0$): the quadratic form of $E(t_0)$ is uniformly bounded. Contradiction. ∎

The problem protects its own equation: the value function bounds E, because E *is* the value function.



The entries of $E(t)$ for the tracking example below: integrated backwards from $E(T) = -W$, bounded on $[0, T]$.

The tracking problem

Make a linear system follow a given curve $\gamma : [0, T] \to \mathbb{R}^n$ (not a trajectory of the system, in general), minimizing the error $z = x - \gamma$ and the control effort.

Proposition 5.3.2 (tracking)
For $\dot{x} = Ax + Bu$, $x(0) = x_0$, with cost

$$J(u) = (x(T) - \gamma(T))^\top W(x(T) - \gamma(T)) + \int_0^T \left[(x - \gamma)^\top R(x - \gamma) + u^\top P u\right] dt,$$

the unique optimal control is the feedback

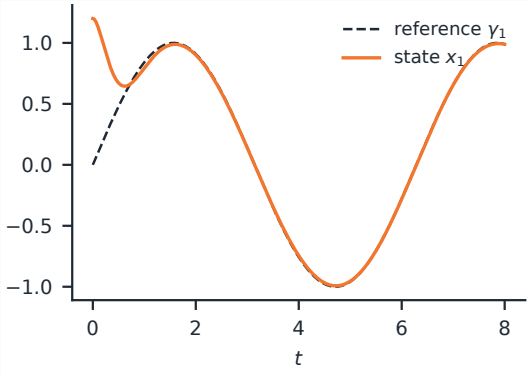
$$u(t) = P^{-1}B^\top E(t)\,(x(t) - \gamma(t)) + P^{-1}B^\top h(t),$$

where E solves the same Riccati equation as before ($E(T) = -W$) and h the linear ODE

$$\dot{h} = -A^\top h - E\,(A\gamma - \dot{\gamma}) - EBP^{-1}B^\top h, \qquad h(T) = 0.$$

Proof idea. The error solves $\dot{z} = Az + Bu + r$ with $r = A\gamma - \dot{\gamma}$. Absorb the drift r by state augmentation, $z_1 = (z, 1) \in \mathbb{R}^{n+1}$; apply the Riccati theorem to the augmented system and split $E_1 = \left(\begin{smallmatrix} E & h \\ h^\top & \alpha \end{smallmatrix}\right)$: the blocks give exactly the two equations above. ∎

Tracking in action, and a summary of Chapter 5



A double integrator tracking a reference curve (dashed): the feedback $u = P^{-1}B^\top E(x - \gamma) + P^{-1}B^\top h$, with E and h integrated backwards from T.

Take-away of Chapter 5

- Optimality = boundary of the *extended* reachable set.
- **PMP**: state + adjoint dynamics + Hamiltonian maximization; it is the *KKT theorem* for the dynamics-as-constraint, with controllability as constraint qualification and transversality from the boundary terms.
- Free terminal state + quadratic penalty: the costate is linear in the state, $p = Ex$, with E solving a **Riccati equation** — optimal control in *feedback* form.
- Tracking reduces to LQR by state augmentation.

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